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Stock Price Prediction Using LSTM:

A Case Study of J.P. Morgan

Math 509

0. Abstract

This project applies a Long Short-Term Memory (LSTM) model to predict stock prices using historical data from J.P. Morgan. Using OHLCV features, the problem is formulated as a time-series forecasting task, where the model generates one-step-ahead predictions. The results show that the model captures overall price trends with relatively low error. However, its performance is limited in predicting sudden changes due to the inherent volatility of financial markets. Overall, historical data is useful for short-term forecasting but cannot fully explain stock price movements.

1. Introduction

Stock price movements are complex and affected by many factors, including market conditions, investor behavior, and external economic events. Due to this complexity, predicting stock prices is a difficult problem. In this project, we analyze and forecast J.P. Morgan's stock price behavior, one of the largest and most influential financial institutions in the United States. With a history of over 200 years and a strong presence in global financial markets, J.P. Morgan serves as a representative example of a large, stable company whose stock reflects broader market dynamics.

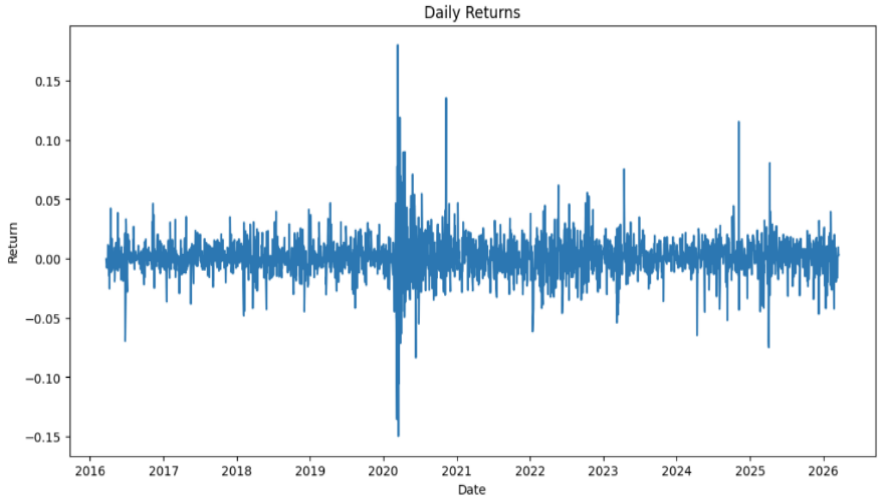
We use historical data from Yahoo Finance, including opening, closing, high, low prices, and trading volume. The goal is to investigate whether past stock data can be used to predict short-term future movements. Specifically, we apply a Long Short-Term Memory (LSTM) model to identify patterns in time-series data and generate predictions. To support this, we conduct exploratory data analysis, including trend, return, volatility, and autocorrelation. The key question is whether historical data can provide reliable short-term predictions and what limitations this approach may have.

2. Exploratory Data Analysis

2.1 Trend Analysis

The closing price of J.P. Morgan shows a clear upward trend over time, indicating overall growth in the stock value. However, there are noticeable fluctuations, including a sharp decline around 2020 followed by a recovery. In recent years, especially after 2023, the stock price has increased more rapidly. This suggests that while the stock exhibits long-term growth, short-term movements remain volatile.

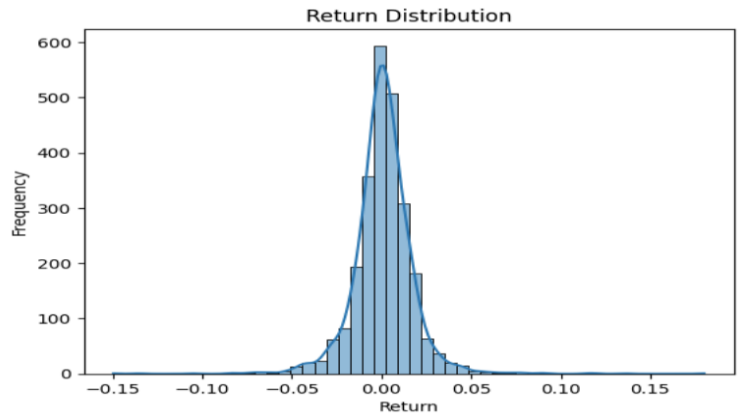
2.2 Return and Volatility Analysis



The daily returns fluctuate around zero, indicating that there is no consistent upward or downward pattern in short-term price changes. Most of the fluctuations are relatively small, but there are occasional large spikes, particularly around 2020, reflecting periods of increased market uncertainty.

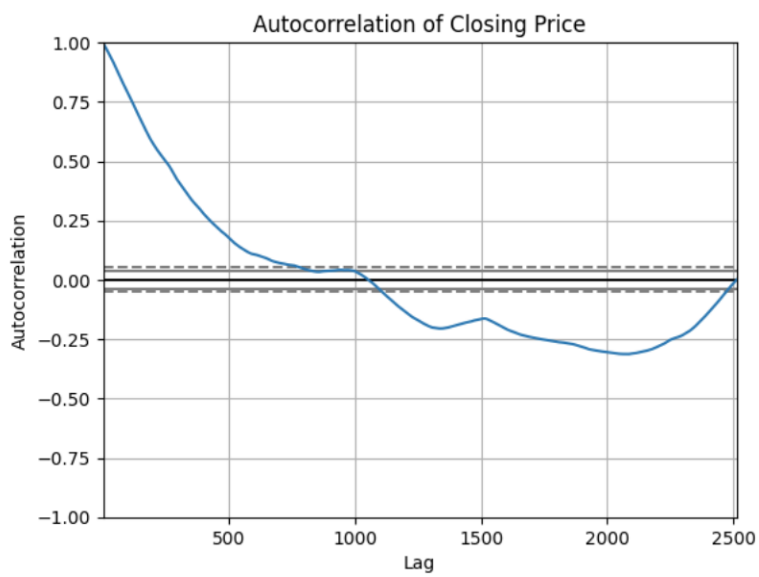
In addition, the magnitude of fluctuations varies over time, with some periods showing more intense movements than others. This suggests that stock returns are volatile and that the level of risk is not constant over time.

2.3 Distribution of Returns



The distribution of daily returns is centered around zero, indicating that most price changes are relatively small. The shape appears approximately symmetric, but with heavier tails compared to a normal distribution, suggesting the presence of occasional extreme movements. This is consistent with periods of market stress, such as the sharp fluctuations observed around 2020, which are reflected in the tails of the distribution. Overall, this indicates that while most returns are moderate, large positive or negative changes can still occur, reflecting the inherent risk in stock price movements.

2.4 Autocorrelation Analysis



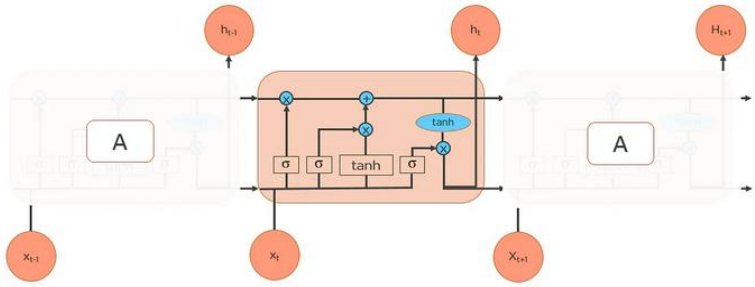
The autocorrelation function shows strong positive correlation at small lags, which gradually decreases as the lag increases. This indicates that current stock prices are highly

dependent on recent past values. As the lag becomes larger, the correlation weakens and eventually approaches zero, suggesting that long-term dependencies are less significant.

These results confirm the presence of temporal dependence in the data, meaning that past observations contain useful information for predicting future values. This justifies the use of sequence-based models such as LSTM, which are designed to capture such time-dependent patterns

3. Methodology

In this project, we formulate stock price prediction as a time-series forecasting problem. Let X_t denote the observed variables at time t, including opening price, closing price, highest price, lowest price, and trading volume. The objective is to use a sequence of past observations X_{t-k}, ..., X_t to predict the future value Y_t, which represents the stock price at time t. This formulation allows the model to capture temporal dependencies in historical data.



To model these sequential relationships, we employ a Long Short-Term Memory (LSTM) network, which is designed to process time-dependent data and retain useful information from previous time steps. The data is organized into sequences of fixed length k, forming a three-dimensional structure consisting of samples, time steps, and features. Each sequence is used as input to predict the value at the next time step.

During training, the model parameters are optimized by minimizing the prediction error, measured using Mean Absolute Error (MAE). To improve generalization and reduce overfitting, we apply techniques such as dropout and early stopping. In addition, the dataset is split chronologically into training and test sets, ensuring that the model is trained on past data and evaluated on future data, consistent with real-world forecasting scenarios.

We assume that historical stock data contains useful information for predicting short-term future price movements, and that patterns observed in the past may persist in the near future. We also assume that the selected variables (open, high, low, close, and volume) are sufficient to capture the main dynamics of stock prices. However, this model has several limitations. It does not account for external factors such as economic news, market sentiment, or unexpected events, which may significantly influence stock prices. Moreover, stock price movements are inherently noisy and may not exhibit stable patterns over time, limiting the model's predictive accuracy.

4. Solution of the Problem

4.1 Data Preparation

To solve the forecasting problem, we first preprocess the historical stock data and transform it into sequential input suitable for time-series modeling. Specifically, the selected features (open, high, low, close, and volume) are standardized to ensure they are on comparable scales, which helps improve model training stability. The data is then organized into overlapping sequences of fixed length, where each sequence contains past observations used to predict the next time step. The dataset is subsequently split into training and test sets in chronological order, ensuring that the model is trained on past data and evaluated on future data.

4.2 Model Implementation

We use a Long Short-Term Memory (LSTM) model to capture temporal dependencies in the stock price data. The model takes sequential input and predicts the value at the next time step, allowing it to learn patterns from historical observations.

The model is trained on the training set by iteratively adjusting its parameters to minimize prediction error. After training, it is used to generate one-step-ahead forecasts on the test set. The performance of the model is evaluated using metrics such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). In addition, predicted values are compared with actual values through visualization to assess how well the model captures short-term fluctuations in stock prices.

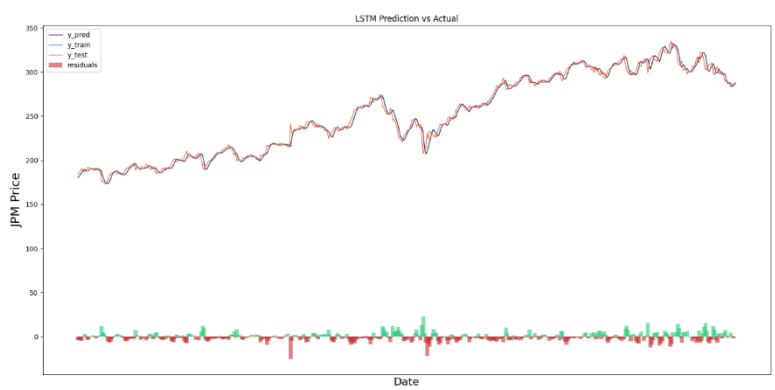
5. Interpretation of results

5.1 Training Performance

The training and validation loss curves show that the model converges quickly during the early epochs. The training loss decreases rapidly and stabilizes at a low level, indicating that the model is able to fit the training data effectively. The validation loss also decreases overall, although it exhibits some fluctuations across epochs.

The gap between training and validation loss remains moderate, suggesting that the model does not suffer from severe overfitting. In addition, early stopping is triggered around epoch 23, with the best model obtained at epoch 18. This indicates that further training does not significantly improve performance and may lead to instability. Overall, the training process demonstrates that the model achieves a reasonable balance between fitting the data and maintaining generalization ability.

5.2 Prediction Performance



The comparison between predicted and actual stock prices shows that the LSTM model is able to closely track the overall movement of the stock price. The predicted values follow the same

trend as the actual values, indicating that the model successfully captures the underlying patterns in the data.

However, there are still small deviations between the predicted and actual values, particularly during periods of rapid price changes. The model tends to smooth out sharp fluctuations and may not fully capture sudden changes. This is also reflected in the residuals, which are generally centered around zero but exhibit larger variations during more volatile periods. Overall, the results suggest that while the model performs well in capturing general trends, its ability to predict short-term extreme movements is limited.

5.3 Quantitative Evaluation

To further assess the model performance, we compute several error metrics. The Mean Absolute Error (MAE) is 3.46, indicating that the average prediction error is relatively small compared to the scale of the stock price. The Mean Absolute Percentage Error (MAPE) is 1.38%, and the Median Absolute Percentage Error (MDAPE) is 1.01%, both suggesting that the model achieves high accuracy in predicting stock prices.

These results are consistent with the visual comparison in the previous section, confirming that the model performs well in capturing the overall trend while maintaining low prediction error.

6. Critique of the Model

Although the LSTM model is able to capture the overall trend of stock prices, it still has several limitations. The model only uses basic features such as open, high, low, close, and volume, which may not fully reflect all factors that influence stock prices. For example, external factors like economic conditions or major news are not included, even though they can have a strong impact on the market. In addition, stock prices are inherently volatile and unpredictable, which makes it difficult for the model to accurately capture sudden changes.

To improve the model, more data and additional features could be included. For example, incorporating more relevant variables may help better capture the factors influencing stock prices. These improvements could enhance the model's ability to learn more complex patterns and improve prediction accuracy.

7. Conclusion

In this project, we used an LSTM model to predict stock price movements based on historical data. The results show that the model is able to capture the overall trend of the stock price and produce reasonably accurate short-term predictions. The error metrics and visual comparisons indicate that the model performs well under relatively stable conditions.

However, the model has limitations in capturing sudden changes and extreme fluctuations, which are common in financial markets. This suggests that while historical data contains useful information for prediction, it is not sufficient to fully explain stock price movements. Overall, the LSTM model provides a useful approach for short-term forecasting, but its predictions should be interpreted with caution.

8. Appendices

https://github.com/louyang-ly/MAT509-Final-Project